TempTrek Environmental Monitoring System

Version 2.5.5 — Complete Installation & System Administration Guide

Document Title:	Installation & Setup Manual	**Target Platform:**	Windows 10 / 11 / Server (x64)
Application Version:	v2.5.5 (Latest Release)	**Required Privileges:**	Administrator (UAC Elevation)
GAMP 5 Category:	Category 4 (Configured COTS)	**Compliance Scope:**	FDA 21 CFR Part 11 / EU Annex 11

WELCOME & QUICK START SUMMARY:
TempTrek v2.5.5 is a desktop environmental monitoring platform designed to record Modbus TCP sensors (temperature, humidity, voltage, power, boolean coils) in real-time. This manual guides you through installation, first-time Super Admin setup, DPAPI encrypted storage, Modbus device mapping, email alerts, and disaster recovery.

1. System Requirements & Compatibility

Before installing TempTrek, verify that the host workstation satisfies the hardware and system specifications:

Component	Minimum Requirement	Recommended Specification
Operating System	Windows 10 x64 (Build 19041+)	Windows 11 x64 or Windows Server 2022
CPU Architecture	x86-64 (64-bit Processor)	x86-64 Multi-core CPU
System Memory (RAM)	2 GB RAM	4 GB RAM or higher
Available Storage	250 MB for binaries & libraries	2 GB+ for multi-year CSV logs & backups
User Access Level	Local Administrator (UAC)	Local Administrator (UAC)
Network Connectivity	Ethernet/Wi-Fi to Modbus LAN	Dedicated Gigabit LAN for Modbus TCP (Port 502)
Runtimes & Drivers	Bundled in Installer (No setup needed)	Python 3.12 (Only if compiling from source)

2. Choosing Your Installation Package

TempTrek v2.5.5 is available in two ready-to-run Windows distribution formats located in the dist/ folder:

- **Native Windows Installer (TempTrek-2.5.5-x64.msi) — Recommended:**
  A standard 64-bit Windows Installer package. Provisions application binaries into C:\Program Files\TempTrek Environmental Monitoring System, creates Start Menu and Desktop shortcuts, and supports Windows Apps uninstall and upgrade management.
- **Standalone Executable (TempTrek Environmental Monitoring System.exe):**
  A single-file portable executable containing Python runtime, Modbus client drivers, and ReportLab tools. Run directly without installation.
- **Inno Setup Executable (TempTrek Environmental Monitoring System_Setup.exe):**
  An alternative setup wizard compiled from installer.iss.

3. Step-by-Step Software Installation

Method A: Installing via Native Windows Installer (.MSI)

1. Open the **dist/** folder and locate **TempTrek-2.5.5-x64.msi**.
2. Right-click the .msi file and select **Install** (or double-click to launch).
3. When the Windows User Account Control (UAC) prompt appears, click **Yes** to grant administrative access.
4. The setup wizard automatically installs binaries to C:\Program Files\TempTrek Environmental Monitoring System.
5. Launch TempTrek anytime from the new **Desktop Shortcut** or from **Start Menu > TempTrek**.

Method B: Deploying Standalone Executable (.EXE)

1. Copy **dist/TempTrek Environmental Monitoring System.exe** to your target directory (e.g. C:\Program Files\TempTrek\).
2. Right-click the file, select **Properties > Compatibility**, and check **Run this program as an administrator**.
3. Double-click the .exe file to start monitoring.

4. First-Time Launch & Super Admin Password Setup

ADMINISTRATOR PRIVILEGES MANDATORY:
TempTrek requires elevated Administrator rights to configure Windows DPAPI encryption and manage backup directory ACL permissions under %ProgramData%\TempTrekBackups. Always right-click and choose **Run as administrator**.

Super Admin Password Creation Rules:

1. On first launch, the software detects an uninitialized configuration and prompts for a master password.
2. **Minimum Password Length:** Must be at least **6 characters** (e.g. len >= 6).
3. **Hashing Standard:** Hashed securely using **salted PBKDF2-SHA256** to prevent credential recovery.
4. **Password Expiration:** Passwords expire after **90 days** by default (configurable to 1, 30, 60, or 90 days).

Application File Storage & DPAPI Encryption Table:

Windows Folder Path	Data Stored	Security Rules & Protection
%APPDATA%\TempTrek\config	devices.json, users.json, email_settings.json, trend_settings.json	Encrypted via **Windows DPAPI** (Data Protection API). Bound to original Windows user account.
%APPDATA%\TempTrek\live	Daily-rotated sensor logs ([IP]_[Port]_s[SlaveID]_[YYYY-MM-DD].csv)	Automated daily log rotation prevents row overflow and keeps data segmented.
%APPDATA%\TempTrek\reports	Generated PDF reports and approval audit records	Contains viewed history, digital approval signatures, and draft watermarks.
%ProgramData%\TempTrekBackups	Password-encrypted .ttbackup archive zip files	Restricted destination. Exclusive ACL permissions granted to **Administrators / SYSTEM**.

5. Modbus TCP & Sensor Channel Configuration

Log in as Super Admin and open **Administration > Node Config** to register sensors:

1. **Add Device Node:** Enter Device IP (e.g. 192.168.1.100), Port (default 502), Slave ID (e.g. 1), and **Time Interval to Store (sec)**.
2. **Register Mapping:** Click *Register Mapping* to map sensor registers:
   - **FC01 (Output Coils)** / **FC02 (Input Coils):** Boolean status signals (e.g. 0 = Normal, 1 = Alarm).
   - **FC03 (Holding Registers)** / **FC04 (Input Registers):** Numerical sensor values (FLOAT32, INT16, INT32, BCD).
3. **Decimal Precision:** Specify channel precision (e.g. 1 or 2 decimals) to format live cards and PDF graphs.
4. **Start Monitoring:** Click **START MONITORING** to launch 1.0-second background acquisition threads.

6. Backup & Disaster Recovery Setup

Open **Administration > Backup & Restore** in an elevated Administrator session:

- **Set Backup Password:** Click *SAVE SETTINGS* or *CREATE BACKUP NOW* to define an archive encryption password.
- **Backup Archive Format:** Backups create password-authenticated .ttbackup zip archives encrypted with AES-GCM.
- **Restoration Safety:** Restoring pauses acquisition, verifies checksums, creates a pre-restore safety snapshot, executes atomic folder replacement with recovery journal rollback, and restarts the app.

7. GAMP 5 Software Classification & GxP Compliance

Under ISPE GAMP 5 (Good Automated Manufacturing Practice, 2nd Edition) guidelines for regulated GxP environments:

GAMP 5 Category	Software Definition	TempTrek System Classification Rationale
Category 4 (Configured)	Commercial Off-The-Shelf (COTS) software configured to fit business requirements without changing code.	**Standard Production Deployment:** Delivered as a compiled package (MSI/EXE). Site administrators configure Modbus IP nodes, register mappings (FC01-FC04), alarm rules, and user roles via standard UI forms.
Category 5 (Custom/Bespoke)	Custom software written from scratch or customized for specific client requirements.	**Source Customization:** Applies if in-house Python source code (2.py, storage_security.py) is modified or extended for custom site protocols.

FDA 21 CFR Part 11 & EU Annex 11 Feature Summary:

- **Audit Trail (11.10e):** Chained SHA-256 audit log tracking user logins, config edits, and viewed data.
- **Access Control (11.10d):** Salted PBKDF2-SHA256 password security, mandatory password change policy, 6+ character length enforcement.
- **Data Security (11.10c):** Windows DPAPI config encryption, exclusive Administrator/SYSTEM backup ACLs, and AES-GCM encrypted `.ttbackup` archives.

8. Post-Installation Verification & Testing

To verify system operational health and security controls on host hardware, run the automated test suite:

```
python -m unittest discover -s tempterk -p "test_*.py" -v
```

All 50 automated tests must return **OK**, validating DPAPI protection, salted PBKDF2 password hashing, atomic disk writes, ACL generation, and restore rollback mechanics.